THE FEDERAL POLYTECHNIC MUBI,

DEPERTMENT OF COMPUTER SCIENCE

WEB TECHNOLOGY (COM 225)

INTERNET: Is a global network of interconnected computers and servers that communicate with each other using standardized protocols, allowing for information sharing, communication, and data exchange.

The internet has six application protocols :Electronic mail(Email), World Wide Web (Hypertext Transfer Protocol or HTTP), Gopher, Telnet, File Transfer Protocol (FTP), Each of these application protocol has special client software, and many web browsers, such as Netscape from communication and internet Explorer from Microsoft, which are capable of reading and displaying data from all the applications. ***Internet enables us to:***

- Access information (websites, online resources)
- Communicate (email, social media, messaging)
- Share data (file transfers, cloud storage)
- Connect devices (IoT, smart devices)

Benefit of Internet

- Global Connectivity: Connect with people worldwide, regardless of geographical distance.
- Information Access: Instantly access vast amounts of information, knowledge, and resources.
- Communication: Facilitate rapid communication through email, social media, messaging, and video conferencing.
- Ecommerce: Shop online, purchase goods and services, and conduct financial transactions.
- Education and Learning: Access online courses, tutorials, and educational resources.
- Entertainment: Stream music, videos, and movies, and play online games.

the fundamentals of the internet:

- TCP/IP: The internet's core communication protocol, allowing devices to connect and exchange data.
- IP Addresses: Unique addresses assigned to devices, enabling them to communicate.

- Packet Switching: Data is broken into packets, transmitted, and reassembled at the destination.
- Networks: Devices connected together, sharing resources and data.
- Protocols: Standardized rules governing data transmission, such as HTTP, FTP, and DNS.
- Servers: Powerful computers storing and managing data, applications, and services.
- Clients: Devices (computers, phones, etc.) accessing servers and internet resources.
- Domain Name System (DNS): Translates domain names into IP addresses.
- Hypertext Transfer Protocol (HTTP): Enables web page requests and responses.
- Internet Service Providers (ISPs): Companies providing internet access and connectivity

Some short definitions

Internet = Global Network

A collection of interconnected devices (computers, servers, smartphones, etc.)

Connected through physical or wireless links (cables, WIFI, etc.)

Devices communicate using standardized languages (protocols)

Key aspects:

Interconnected: Devices can talk to each other

Global: Spanning the entire world

Network: A web of connected devices

***The World Wide Web* (WWW)**

Is a system of linked documents (web pages) accessed via the internet and way to share and view information using web browsers and hyperlinks and also global network of interconnected documents and resources.

It can also be defined as a system of hypertext document and other resources that are accessible over the internet.

What is the WWW?

- WWW stands for the World Wide Web
- The World Wide Web is most often called the Web
- The Web is a network of computers all over the world

- All the computers in the Web can communicate with each other
- All the computers use a communication standard called HTTP

WWW.Web Basic

- URL
- HTTP
- WEB BROWSER

URL: URL stands for Uniform Resource Locator, which is a web address used to locate a specific resource on the internet. A URL typically consists of:

1. Protocol (http or https)

2. Domain name

3. Path (e.g., /about/team)

4. Query string (e.g., name=John&age=30)

5. Fragment (e.g., #top)

HTTP: HTTP stands for Hypertext Transfer Protocol, which is the foundation of the web. It's a protocol that allows devices to communicate with each other and exchange information over the internet.

Web browser: A web browser is a software application that allows users to access and view websites on the internet. Here are some key aspects of web browsers:

Functions:

i. Rendering: Displays web pages on the screen
ii. Navigation: Allows users to move between web pages
iii. Searching: Finds web pages using search engines
iv. Bookmarks: Saves favorite web pages for easy access
v. History: Keeps a record of visited web pages

WEB development

Web refers to the website, webpage or anything that work over the internet.

Web development is the process of building and maintaining websites, web applications, and mobile applications. *It involves various tasks, such as:*

- Frontend development: Creating user interfaces, layouts, and visual effects using HTML, CSS, JavaScript, and frameworks like React, Angular, or Vue.

- Backend development: Building server-side logic, database integration, and API connectivity using languages like Java, Python, Ruby, PHP, and frameworks like Node.js, Django, or Ruby on Rails.
- Full stack development: Combining frontend and backend development skills to create complete web applications.

Web Development Process

- Plan: Define purpose, content, and structure.
- Design: Create visual layout and user experience.
- Build: Write HTML, CSS, and JavaScript code.
- Test: Ensure functionality and usability.
- Launch: Publish website online.

WEB development frame work application

React: is use with libraries like react players for embedding

Vue.JS:

Angular:

Key Components of web applications

1. HTML (Hypertext Markup Language): Structuring content.

2. CSS (Cascading Style Sheets): Styling and layout.

3. JavaScript: Adding interactivity.

1. HTML (Hypertext Markup Language): is a web programming language designed to create web documents or web pages. It's the standard markup language used to create web pages and display content on the internet.

HTML is like a set of instructions that tells a web browser how to structure and display content, such as: Headings, Paragraphs, Images, Links, Forms, Tables, Lists

HTML uses tags (elements) to wrap around content and apply meaning to it. For example:

`<p>This is a paragraph</p>` defines a paragraph

`<img src="image.jpg">` displays an image

`<a href="(link unavailable)">Visit Example</a>` creates a hyperlink

HTML is the backbone of a website, providing the structure and content that the web browser renders to the user.

Some key aspects of HTML:

Markup language: uses tags to add meaning to content

Standardized: follows W3C (World Wide Web Consortium) standards

Platform independent: works on various devices and browsers

Evolving: new features and elements are added over time (e.g., HTML5)

Scripting in HTML

Scripting in HTML is typically done using JavaScript, which is a programming language that allows you to add interactivity and dynamic effects to your web pages.

Here are some ways to add scripting to HTML:

1. Inline Scripting: Add JavaScript code directly to an HTML element using the `onclick`, `on mouseover`, etc. attributes.

2. Internal Scripting: Add JavaScript code within the `<script>` element inside the HTML document.

3. External Scripting: Link to an external JavaScript file using the `src` attribute of the `<script>` element.

Some common scripting tasks in HTML include:

Event handling: Responding to user interactions like clicks, hover, etc.

DOM manipulation: Dynamically changing HTML content, styles, or structure.

Form validation: Checking user input data for errors or validity.

Animations and effects: Creating interactive visual effects.

2. CSS (Cascading Style Sheets): CSS (Cascading Style Sheets) is a styling language used to control the layout and appearance of web pages. It's used to separate presentation from structure, making HTML more focused on content and semantics.

CSS consists of:

- Selectors: Identify HTML elements to apply styles to.
- Properties: Define the styles to apply (e.g., color, font size, margin).
- Values: Specify the values for properties (e.g., #ff0000, 18px, 10px).

CSS concepts:

- Box model: Elements are represented as boxes with margin, border, padding, and content.
- Layout modes: Block, inline, flexbox, grid, and table layouts.
- Positioning: Static, relative, absolute, fixed, and sticky positioning.
- Display: Control element visibility and type (block, inline, none).
- 10. Borders: Control border width, style, and color.

Some popular CSS tools and resources include:

- CSS preprocessors: Sass, Less, and Stylus.
- CSS frameworks: Bootstrap, Tailwind CSS, and Bulma.
- CSS linters: Style lint and CSS Lint.
- CSS formatters: Prettier and CSS Formatter.
- CSS documentation: MDN Web Docs and CSS Tricks.

DINAMIC CONSTANT

In programming, a dynamic constant refers to a value that is:

- Initialized at runtime: Its value is determined when the program runs, not at compile time.
- Constant: Once initialized, its value cannot be changed.

Dynamic constants are useful when:

- Values are calculated: Based on user input, environment variables, or other factors.
- Values are retrieved: From a database, file, or external source.
- Values are generated: Using algorithms or random number generators.

Examples of dynamic constants include:

- User defined settings: Stored in a database or file.
- Environment variables: Set by the operating system or user.
- Calculated values: Based on user input or other dynamic data.
- Random values: Generated using a random number generator.

Understanding JavaScript

JavaScript is used in millions of Web pages to improve the design, validate forms, detect browsers, create cookies, and much more.

JavaScript is a high-level, dynamic, and interpreted programming language that is primarily used for client-side scripting on the web. ***It's commonly used for:***

- Web development: Creating interactive web pages, web applications, and mobile applications.
- Frontend development: Building user interfaces, layouts, and visual effects.
- Backend development: Building server-side logic, database integration, and API connectivity with technologies like Node.js.
- Desktop and mobile app development: Creating desktop and mobile applications using frameworks like Electron and React Native.
- Game development: Building games using frameworks like Phaser.

Features

- Dynamic: JavaScript is dynamically typed, meaning you don't need to declare variable types before using them
- Interpreted: JavaScript code is executed line by line, without the need for compilation
- Object-oriented: Supports object-oriented programming (OOP) concepts like inheritance, polymorphism, and encapsulation
- First-class functions: Functions are treated as first-class citizens, meaning they can be passed as arguments, returned as values, and stored in data structures
- Prototype based: JavaScript uses prototypes instead of classes for inheritance

Advantages

- Dynamic typing: JavaScript variables can change type at runtime
- Flexibility: JavaScript variables can hold any type of value
- Easy to declare: variables can be declared with minimal syntax (e.g., let x;)

Operators

1. Assignment operators

2. Comparison operators

3. Arithmetic operators

4. Logical operators

5. Bitwise operators

RELATIONSHIP BETWEEN CSS, HTML AND JAVASCRIPT

CSS, HTML, and JavaScript are three fundamental technologies that work together to create web pages and web applications. Here's a brief overview of their relationship:

1. HTML (Hypertext Markup Language): Provides the structure and content of a web page.

2. CSS (Cascading Style Sheets): Controls the layout, visual styling, and user experience of a web page.

3. JavaScript: Adds interactivity, dynamic effects, and functionality to a web page.

Relationship:

HTML provides the foundation, CSS enhances the presentation, and JavaScript adds behavior.

HTML elements are styled using CSS selectors and properties.

JavaScript manipulates HTML elements and their CSS styles to create dynamic effects.

JavaScript events are triggered by user interactions with HTML elements.

CSS animations and transitions can be triggered by JavaScript.

TOGETHER, THEY ENABLE:

Responsive web design: Adapt to different screen sizes and devices.

Interactive web pages: Engage users with dynamic effects and functionality.

Web applications: Create complex, data driven applications.

In summary, HTML provides the structure, CSS enhances the presentation, and JavaScript adds interactivity, making them a powerful trio for building web experiences.

DIFFERENCE BETWEEN CSS, HTML, AND JAVASCRIPT

Here are the main differences between CSS, HTML, and JavaScript:

CSS (Cascading Style Sheets)

Purpose: Styling, layout, and visual effects

Focus: Presentation and user experience

Syntax: Declarative (rules based)

Functionality:

- Layout management
- Visual styling
- Animations and transitions
- Responsive design

HTML (Hypertext Markup Language)

Purpose: Structuring and content definition

Focus: Content and structure

Syntax: Markup based (tags and attributes)

Functionality:

- Defining content and structure
- Creating hyperlinks and navigation
- Embedding multimedia and resources
- Semantic meaning

JavaScript

Purpose: Dynamic behavior and interactivity

Focus: Functionality and interaction

Syntax: Imperative (code based)

Functionality:

- Dynamic effects and animations
- User interaction and event handling
- Data manipulation and storage
- Server-side communication (AJAX)

In summary:

HTML defines the structure and content

CSS enhances the presentation and layout

JavaScript adds dynamic behavior and interactivity

WEB DEVELOPMENT TOOLS

Here are some popular web development tools.

- Code Editors
- Integrated Development Environments (IDEs)
- Version Control Systems
- Debugging Tools
- Testing Frameworks
- Build Tools
- Package Managers
- Design and Prototyping Tools
- Accessibility Tools
- Security Tools

MULTIMEDIA

Multimedia refers to content that uses multiple forms of media, *such as:*

- Text: Written words, like articles, captions, and subtitles.
- Images: Still pictures, like photos, graphics, and illustrations.
- Audio: Sound files, like music, voiceovers, and sound effects.
- Video: Moving images, like movies, animations, and live streams.
- Interactive elements: User input elements, like buttons, forms, and games.

Multimedia is used in various fields, including:

1. Education: Online courses, tutorials, and educational software.

2. Entertainment: Movies, video games, and interactive stories.

3. Marketing: Advertisements, product demos, and promotional videos.

4. Communication: Video conferencing, online meetings, and social media.

UNDERSTANDING XML

XML (Extensible Markup Language) is a markup language used to store and transport data in a format that is both human readable and machine-readable. Here's a breakdown of understanding XML:

Basic Concepts:

1. Elements: Represented by tags, which are surrounded by angle brackets (< >). Elements can have attributes and contain text or other elements.

2. Tags: Used to define the start and end of an element. Opening tags have a less than sign (<) and closing tags have a greater than sign (>).

3. Attributes: Added to elements to provide additional information. They consist of a name and value, separated by an equal's sign (=).

4. Content: The text or other elements contained within an element.

XML Structure:

1. Root Element: The topmost element, which contains all other elements.

2. Child Elements: Elements contained within another element.

3. Parent Element: The element that contains another element.

4. Siblings: Elements that share the same parent element.

XML Uses:

1. Data Exchange: XML is used to exchange data between systems, applications, and organizations.

2. Configuration Files: XML is used to store configuration data for applications and systems.

3. Data Storage: XML can be used to store data in a format that is both human readable and machine-readable.

XML Tools:

1. Parsers: Software that reads and interprets XML documents, such as SAX and DOM.

2. Validators: Tools that check XML documents against a DTD or XML Schema, such as Xerces and MSXML.

3. Editors: Software that allows users to create and edit XML documents, such as XMLSpy and Oxygen XML Editor.

TEST ANSWERS / SUMMARY

1. **WEB SARVER**: is a computer that stored web page. Or a web saver is a tool that helps you safe and organize web page or content.
2. **WEBSITE**: is a one or more web page belonging to a particular company, institute, or organization or an individual.
3. **WEB PAGE**: is an electronic document written in a computer language called HTML

CSS (Cascading Style Sheets): is a simple design language intended to simplify the process of making webpage presentable.

ADVANTAGE OF CSS

Save style

Page load faster

Easy to maintain

Compatible with multiple devices

METHOD OF CONNECTIN HTML WITH CSS

i. Inline Style Sheet
ii. Internal Style Sheet
iii. External Style Sheet

DNS (domain name Sarver): is a unique name of an individual host computer in the internet

JAVA SCRIPT: is a light weight object-oriented programming language that is primarily used for client-side scripting on the web.

ADVANTAGE OF JAVASCRIPT

1. Less saver interaction
2. Increase interactivity
3. Reacher interface
4. Immediate feedback to the user

INTERNET: is an interconnected of large and small computers network around the globe.

MULTIMEDIA: Multimedia refers to content that uses multiple forms of media, *such as:* Text, Images, Audio, videos and the are also the components.

TOOLS AND TECHNOLOGY OF MULTIMEDIA

Images edition and optimization***:*** e.g. photoshop, photo lab, adobe etc.

Audio edit: like adobe audition, etc.

Video edit: like snap-chart, cap-cut etc.

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Amination and interactive media: like adobe animate

Video: Moving images

HOW TO CREAT A HTML TABLE WITH SOME CONTENT IN IT

```
<table boarder =”2” cell padding =”2”>
<tr>
<th> 8-10</th>
<th> 10-12</th>
<th> 12-2</th>
</tr>
<tr>
<td>com 111 </td>
<td>com 112 </td>
<td>com 113 </td>
</tr>
<tr>
<td> com 114 </td>
<td> com 115 </td>
<td> com 116 </td>
</tr>
</table>
```

8-10	**10-12**	**12-2**
COM 111	COM 112	COM 113
COM 114	COM 115	COM 116

THE OUTPUT

HOW TO CREATE A HEADER / TITLE NAME IN HTML

```
<!Doctype>
<HTML>
<head>
<title>com225</table>
</head>
</body>
```